MATH3900

Tutorial 4

1. Let $f : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable, μ-strongly convex and L-smooth, where $0 < \mu \le L$. Consider the following algorithm with zero stopping tolerance.

Newton's method with Armijo backtracking

Input: $x_0 \in \mathbb{R}^d$, $\bar{\alpha} > 0$, and $c_A, \eta \in (0, 1)$.

for $k = 0, 1, 2, \ldots$ **do**
  Compute $g_k = \nabla f(x_k)$.
  if $g_k = 0$ **then**
    stop and return x_k.
  end if
  Compute $H_k = \nabla^2 f(x_k)$; solve $H_k p_k = -g_k$.
  Set $\alpha = \bar{\alpha}$.
  while $f(x_k + \alpha p_k) > f(x_k) + c_A \alpha g_k^T p_k$ **do**
    Set $\alpha = \eta\alpha$.
  end while
  Set $\alpha_k = \alpha$ and $x_{k+1} = x_k + \alpha_k p_k$.
end for

You may use the Hessian bounds $\mu I \preceq H_k \preceq LI$. The global convergence theorem proved in class may be used without proof: for a bounded-below, L-smooth objective, the uniform bounds

$$g_k^T p_k \le -\sigma \|g_k\|^2, \qquad \|p_k\| \le \beta \|g_k\|$$

with fixed $\sigma, \beta > 0$ imply $\|g_k\| \to 0$.

Prove that this algorithm converges to the unique minimizer $x^\star$ from any starting point, and that the additional assumptions in part (d) give eventual quadratic convergence.

(a) **Uniform direction bounds.** Show that Newton's direction satisfies the theorem's bounds with $\sigma = 1/L$ and $\beta = 1/\mu$.

(b) **Distance to the minimizer.** Using strong convexity at x and $x^\star$, show that

$$\|x - x^\star\| \le \frac{\|\nabla f(x)\|}{\mu}.$$

(c) **Global convergence.** Apply the lecture's theorem and part (b) to prove that Newton with backtracking converges to $x^\star$ from any starting point.

(d) **Eventual full steps.** Suppose additionally that $\bar{\alpha} = 1$, $c_A < 1/2$, and the Hessian is ρ-Lipschitz, with $\rho \ge 0$, in a neighborhood of $x^\star$:

$$\left\|\nabla^2 f(u) - \nabla^2 f(v)\right\|_2 \le \rho \|u - v\|$$

for u, v in that neighborhood. At a point x, write $g = \nabla f(x)$, $H = \nabla^2 f(x)$, and let $Hp = -g$. If the full-step segment stays in this neighborhood, you may use

$$f(x+p) \le f(x) + g^T p + \tfrac{1}{2} p^T H p + \tfrac{\rho}{6} \|p\|^3 .$$

Show that the full step satisfies Armijo whenever

$$\tfrac{\rho}{6} \|p\| \le \mu(1/2 - c_A).$$

Deduce that the algorithm eventually accepts $\alpha_k = 1$.

(e) **Quadratic convergence.** Under the additional assumptions in part (d), prove that, for all sufficiently large k,

$$\|x_{k+1} - x^\star\| \le \frac{\rho}{2\mu} \|x_k - x^\star\|^2 .$$

Explain what this inequality says about the error in successive iterations. *Hint:* with $e = x - x^\star$ and $H(z) = \nabla^2 f(z)$, the fundamental theorem of calculus gives

$$\nabla f(x) = \int_0^1 H(x^\star + te) e \, dt.$$

2. Determine whether each statement is *True* or *False*. Give a short proof for a true statement, or a counterexample with verification for a false statement.

   (a) For fixed c_A, backtracking always returns the largest positive step size satisfying Armijo.

   (b) If $f : \mathbb{R} \to \mathbb{R}$ is bounded below and has a Lipschitz continuous derivative, every point with $f'(t) = 0$ is a global minimizer.

3. Let

$$f(x) = x_1^2 + 3x_2^2, \qquad x = (1,1)^T, \qquad p = -\nabla f(x).$$

Use Armijo backtracking with $\bar{\alpha} = 1$ and $\eta = 1/2$.

   (a) Compute p, $\nabla f(x)^T p$, and $\phi(\alpha) = f(x + \alpha p)$.

   (b) Take $c_A = 1/4$. Check the trials in order until the first acceptance. For each trial, give the objective value and the Armijo bound. Find the accepted point.

   (c) Find the exact interval of acceptable positive steps for $c_A \in (0,1)$. Use it to find the first accepted trial when $c_A = 3/4$. Give a rejected trial that nevertheless decreases f.